



STANDARD COLLEGE NTUNGAMO

S6 PHYSICS 2026 TEST 2

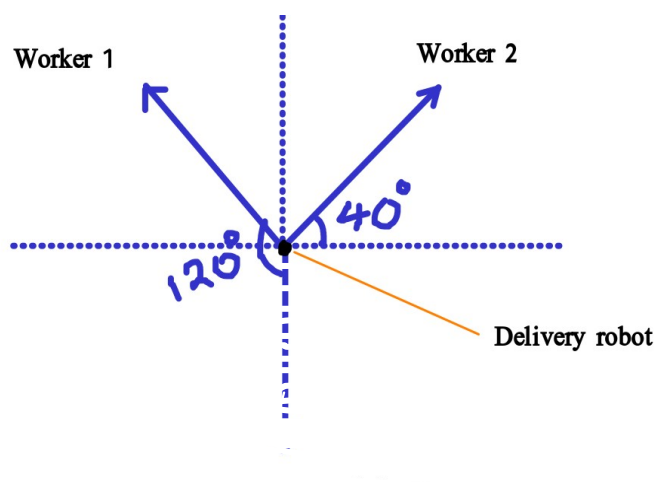
Time: 1 hour

ITEM

During a practical training session in Kampala, a technician is assigned multiple tasks within the same work environment. He begins in the warehouse, where he operates a small delivery robot of mass **0.70 kg** used to transport materials. As it moves, one worker (Worker 1) pushes it forward with a force of **8.0 N** while another (Worker 2) pulls it with a force of **12.5 N**, and friction from the floor opposes the motion and the coefficient of friction is **0.4** as shown below.

After completing this task, the technician proceeds to the garage section where he needs to loosen a tightly fixed bolt on a truck wheel using available tools. He considers using either a short spanner or a long spanner. Once the bolt is removed, he tries to open a tightly sealed jar by applying equal and opposite forces with both hands on the lid.

Later in the day, the technician is sent to a nearby site in Ntungamo to assist a painter. The painter has placed a **10 m** ladder of weight **400 N** against a smooth wall, with its foot resting on rough ground at an angle of **60°** to the ground. The ladder **just slips** when the painter of weight **800 N** climbs to a point **6 m** up the ladder.



TASK:

- Calculate its acceleration as it moves across the warehouse floor.
- Explain which spanner the mechanic should prefer when loosening the bolt.
- Explain what is meant by a **couple** and describe its effect on the jar lid.
- Explain how the forces acting on the ladder keep it in equilibrium.
- Calculate the reaction on the wall and the ground

(f) What is the distance the technician of weight 600N can climb up the ladder so that the same reactions are exerted.

END